Advances in Algorithms (CS503)

Mid Semester Examination, September 2017

Department of CSE, IIT Patna

Total Marks: 60 Duration: 120 min

1. Answer the following questions: [6+6+3]

   I. Prove the correctness of greedy algorithm for Interval scheduling problem.
   II. Show that (S, I_k) is a matroid, where S is any finite set and I_k *is* the set of all subsets of S of size at most k, where $k \leq |S|$.
   III. What are the two properties of any optimization problem that motivate for a greedy algorithm for that problem? Describe them very briefly (not more than 2 lines for each).

2. Answer the following questions: [3 +4+8]

   I. What is the similarity between greedy and dynamic programming approach? Give an example of two closely related problems where they differ.
   II. Write a dynamic programming algorithm for 0-1 knapsack problem.
   III. Give a branching strategy and a bounding strategy for solving Travelling Salesman problem using Branch and Bound approach.

3. Answer the following questions: [7 +4+4]

   I. A full dynamic array creates a new array of twice size and copy all the existing elements in the new array when an append operation is called. Now consider a dynamic array initialized with size 1 and a sequence of m append operations on it. Analyze the amortized cost per operation. Here m is unrestricted.
   II. Describe how to generate Residual graph. Why do we use augmenting path in solving Max-flow problem?
   III. Using both ways reduction show that vertex-cover problem and independent-set problem are equivalent.

4. Answer the following questions: [4+4+7]

   I. Write a poly time algorithm for independent set problem in a tree.
   II. Define Approximation ratio, PTAS and FPTAS.
   III. What is LPT-rule? Show that LPT-rule is a 3/2-approximation algorithm for Load Balancing problem.